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Introduction to AFFDEX SDK 5.1

This article describes Affectiva's AFFDEX SDK 5.1, released February 2022.

If you are using AFFDEX SDK 4.0, please refer to this article for an explanation of the channels and data.

AFFDEX 5.1 includes several critical improvements over SDK 4.0, including increased trackability of faces in suboptimal conditions; increased accuracy in detecting action units; and two new emotion metrics. Future version of iMotions will integrate version 5.1 by default, but if you are in the middle of a study using 4.0, it is advised that you finish data collection with that SDK before upgrading. Please contact your Customer Success Manager for more information.

The Training Dataset

As of 2022, Affectiva's foundational database over the past twelve years has accumulated a total of 12 million face videos from 90 different countries; a total of over 6 billion facial frames.

Of this primary database, 55,000 videos were then used to build AFFDEX 5.1. Each video contains a unique face for a total of 6.5 million frames. The breakdown of the 55,000 videos is as follows:

- 49,500 videos were used to train the algorithms
- 5,900 videos were used for validating the algorithms
- 8,200 videos were used for testing and determining accuracy.

This subset of the data was specifically chosen to build the SDK to avoid overtraining in the most common action units (smile, brow furrow) and overrepresentation for any one ethnicity or age group. These 55,000 faces were picked to produce the best representation and the greatest diversity of expressions and ethnicities across the entire human race. This has resulted in significant improvements in accuracy for certain demographics, with better recall especially for older African & South Asian populations, and better representation for individuals under 18.

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Similar to SDK 4.0, all ground truth data is analyzed by at least 3 FACS-certified human coders to ensure consistency with the academic standards of the Facial Action Coding System. There must be at least 50% consensus between coders on the presence of a particular action unit or expression before the video can be deemed acceptable for training the SDK.

Face Detection & Classification

Improved Face Trackability & Changes to Landmarks

The face tracker function has been overhauled in 5.1, improving the ability of the SDK to detect the face by 16%. The greatest improvements are for "in the wild collections" that SDK 4 has historically struggled with:

- low light and backlit environments
- videos recorded using a mobile phone
- recordings taken from farther away (at least 40px interocular distance)
- washed-out faces
- infrared light recordings

The head angle tolerance has also improved, from 25 degrees to 40.

Since the algorithms to detect the face have changed, the number of facial landmarks used has decreased from 32 to 4. If your study is heavily reliant on the raw facial landmark coordinate data, it is advised that you keep using SDK 4.0.

Action Unit Calculation & Improvements in Accuracy

AFFDEX 5.1, as well as 4.0, provides scores for 20 Action Units as defined by the Facial Action Coding System, as well as Attention:

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Attention

Measure of focus based on the head orientation



Brow Furrow

Both eyebrows moved lower and closer together



Brow Raise

Both eyebrows moved upwards



Cheek Raise

Lifting of the cheeks, often accompanied by "crow's feet" wrinkles at the eye corners



Chin Raise

The chin boss and the lower lip pushed upwards



Dimpler

The lip corners tightened and pulled inwards



Eye Closure

Both eyelids closed



Eye Widen

The upper lid raised sufficient to expose the entire iris



Inner Brow Raise

The inner corners of eyebrows are raised



Jaw Drop

The jaw pulled downwards



Lid Tighten

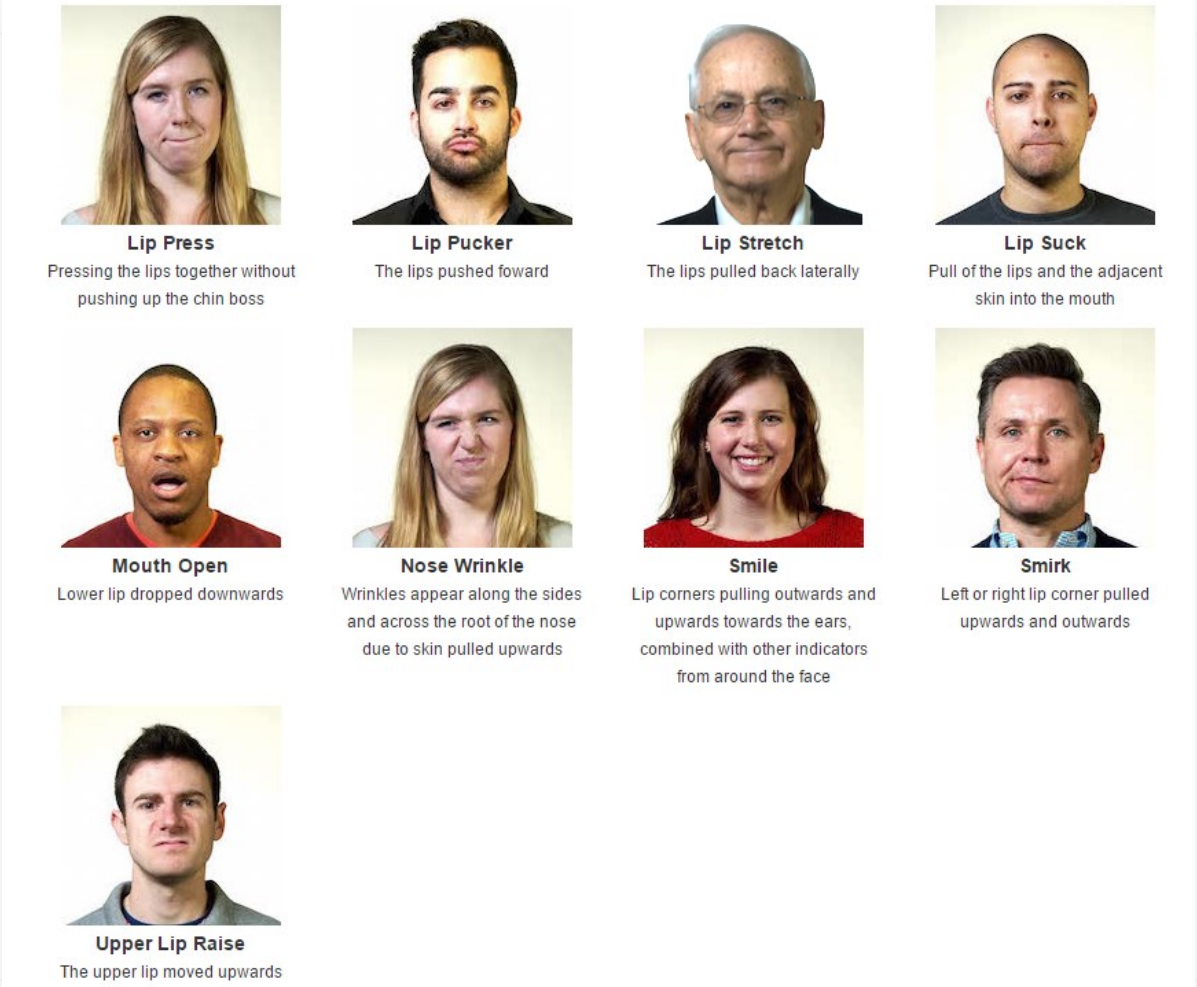
The eye aperture narrowed and the eyelids tightened



Lip Corner Depressor

Lip corners dropping downwards (frown)

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The AU probabilities were calculated by a combination of support vector machine and recursive convoluted neural networks (CNNs).

AFFDEX 5.1 has completely rebuilt the 20 AU classifiers from the ground up using CNNs, thus basing the entire SDK on deep learning.

This has resulted in the majority of AUs showing increased accuracy, particularly for more obscure AUs like Lip Pressor and Cheek Raise. The average ROC-AUC for the new classifiers overall is 91%.

Changes to Baselining Options

AFFDEX has historically not required a baseline neutral stimulus because the classifiers employ an internal "causal" or "rolling" baseline. When the rolling baseline is active, each frame being analyzed is also compared to the faces made in past and future frames, in an effort to reduce false positives by subtracting prolonged or persistent resting faces.

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The assumption behind the rolling baseline is that true, natural facial expressions are dynamic events that manifest and disappear over time, thus persistent expressions are "not natural" and filtered out of the calculation. Internal research has shown that the use of the rolling baseline provides increased accuracy in detecting AUs.

In SDK 4.0, it is not possible to shut off the rolling baseline. With SDK 5.1, users have the option to toggle the baseline as "on" or "off". When baselining is turned "off", each video frame is evaluated in isolation.

High-Level Emotion Scores & New Metrics

Valence & Engagement (Expressiveness)

Valence and Engagement continue to be derived from combinations of action unit scores. These variables will naturally be improved by the increased performance in the base AU classifiers.

Valence consists of the following:

- Inner Brow Raise
- Brow Furrow
- Cheek Raiser
- Nose Wrinkler
- Lip Corner Depressor
- Chin raiser
- Lip Pressor
- Smile

Lip Suck & Upper Lip Raise were previously used in SDK 4.0, now removed from SDK 5.1.

Increase Positive Likelihood	Increase Negative Likelihood
Smile Cheek Raise	Inner Brow Raise Brow furrow Nose wrinkle Lip corner depressor Chin raise Lip press

Engagement consists of the following:

- Inner Brow Raise
- Brow Raise
- Brow Furrow

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- Cheek Raiser
- Nose Wrinkler
- Lip Corner Depressor
- Chin Raiser
- Lip Pressor
- Mouth Open
- Lip Suck
- Smile

Inner Brow Raise is a new addition between SDK 4.0 and 5.1.

How do we calculate engagement?

Engagement, also referred to as Expressiveness, is defined as a measure of facial muscle activation that illustrates the subject's emotional engagement. The range of values is from 0 to 100. Engagement is a weighted sum of the following facial expressions:

- Inner and outer brow raise.
- Brow furrow.
- Cheek raise.
- Nose wrinkle.
- Lip corner depressor.
- Chin raise.
- Lip press.
- Mouth open.
- Lip suck.
- Smile.

Data Export Legend

To export data, right-click the study or analysis, and select "Export" > "Sensor Data".

Sensor_data_default_affectiva.png

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The channels described in section 3 of this article can be exported as the sensor data for Affectiva AFFDEX. With the exception of valence, and the head/eye position information, all of these channels will have values between 0 and 100 representing the probability the metric was expressed (0-it was not expressed, 100-it was expressed, 50-50% likelihood that it was expressed). Note that for these ranges, a threshold of 50 can be set to create binary streams indicating expression/no expression.

Below is a subset of metrics exported from Sensor Data.

Expression & Range	Description	Interpretation
Smile (0 - 100)	Lip corners pulling outwards along with other cues from the face (e.g. eyes) combine to indicate a true smile.	A natural, positive smile. Normally a sign of enjoyment.
Brow Furrow (0 - 100)	Both eyebrows moving lower and closer together.	Can indicate concentration, confusion or dislike.
Brow Raise (0 - 100)	Both eyebrows move upward.	Spontaneously occurs during surprise.
Lip Corner Depressor (Frown) (0 - 100)	Lip corners dropping downward (frown).	Often associated with sadness.
Valence (-100 - 100)	A measure of the positive (or negative) nature of the participant's experience.	-100 negative effect 0 neutral effect 100 positive effect
Engagement (0 - 100)	A measure of emotional responsiveness triggered by the content, whether positive or negative.	Increasing values signify increased emotional engagement.
Attention (0-100)	Measure of point of focus of the user based on the head position	Increasing values signify increased focus.

7 Core Emotions

AFFDEX SDK 5.1 continues to provide scores for the 7 Core Emotions: Joy, Anger, Fear, Surprise, Sadness, Contempt and Disgust, as well as introducing Sentimentality, and Confusion (these latter two are explained further below):

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Anger



Sadness



Disgust



Fear



Joy



Surprise



Contempt

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Emotion	Increase Likelihood	Decrease Likelihood
Joy	Cheek Raise Smile	/
Anger	Brow Furrow Eye Widen Lid Tighten Upper Lip Raise Chin Raise Lip Press Mouth Open	Inner Brow Raise Outer Brow Raise Nose Wrinkle Smile
Disgust	Nose Wrinkle Upper Lip Raise Chin Raise Mouth Open	Inner Brow Raise Smile
Surprise	Inner Brow Raise Outer Brow Raise Eye Widen Jaw Drop	/
Fear	Inner Brow Raise Outer Brow Raise Brow Furrow Eye Widen Lip Stretcher Mouth Open	Nose Wrinkle Smile
Sadness	Inner Brow Raise Brow Furrow Cheek Raise Lip Corner Depressor Chin Raise	Outer Brow Raise Nose Wrinkle Smile
Contempt	Brow Furrow Smirk	Outer Brow Raise Smile
Sentimentality	Inner Brow Raise Brow Furrow Cheek Raise Lip Corner Depressor Chin Raise Lip Press Lip Suck Smile	Outer Brow Raise Mouth Open
Confusion	Inner Brow Raise Brow Furrow Lid Tighten Nose Wrinkle Dimpler Lip Corner Depressor Lip Pucker Jaw Drop	Smile

The Emotion scores are still derived from combinations of action unit scores, but these combinations have been overhauled in SDK 5.1 to align more closely with the definitions described in Ekman & Friesen's EMFACS-7. [1] These variables will naturally be improved by the increased performance in the base AU classifiers. In particular, less commonly shown emotions like contempt and sadness are detected with greater confidence and sensitivity.

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New Metrics: Confusion & Sentimentality & Blink Statistics

AFFDEX 5.1 presents two new high-level emotion scores and two new expressive metrics.

Sentimentality is defined as "happy sadness", and is often seen when watching emotionally resonant material. To isolate the AUs that can be associated with sentimentality, Affectiva presented a battery of advertisements that were thought to elicit sentimental responses to viewers, who ranked the videos on their level of perceived sentimentality.

The most sentimental ads as defined by self-report were then shown to a larger testing community in the wild while recording their facial expressions over video. Gesture analysis was then conducted to determine what combinations of AUs were most associated with a "Sentimental" reaction. Contrary to other Emotion scores, Sentimentality has several possible combinations of AUs, commonly seen together, that can contribute to a positive score.

Confusion was created in an effort to parse different contexts for the use of Brow Furrow. Brow Furrow can be suggestive of Anger and negative valence in general, but can also suggest mental effort or confusion. AFFDEX's Confusion metric attempts to isolate a particular set of expressions that suggest not only mental workload, but also a *lack of comprehension.*

While Confusion is not officially part of the EMFACS system, there have been attempts in the scientific literature to qualify the AU combinations associated with Confusion [2]. Here, Affectiva employed a hybrid approach based on what's been published on prior literature, but also running their own in-house gesture analysis to define the AUs associated with Confusion.

These AUs can include: Brow furrow, Inner brow rise, Lid tighten, and Lip Corner Depressor.

The Blink is detected when the eye closure score gets above and then below a certain threshold during a specific time range. Second, the Blink Rate is calculated on the top of the blink output, and it is the number of blinks happening per minute.

Note: The Affdex Scores (described in this article) will have values between 0 and 100 representing the probability the metric was expressed (0=it was not expressed, 100=it was expressed, 50=50% likelihood that it was expressed).

References

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[1] Friesen, W. V., & Ekman, P. (1984). EMFACS 7.

[2] D'Mello, S.K., Craig, S.D., Graesser, A.C. (2009). Multimethod assessment of affective experience and expression during deep learning. *International Journal of Learning Technology*

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